Majorana Modes in the Machine

Project Summary — Kitaev Chain & Majorana Zero Modes on NISQ Hardware

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The Project in One Slide

The bridge: fermions → qubits → NISQ

- The **1D Kitaev chain** is the minimal model of a 1D topological superconductor: in the topological phase it hosts **Majorana zero modes** localized at its two ends.
- A **Jordan–Wigner** map turns that fermionic model into an exact qubit Hamiltonian — so a quantum device can, in principle, host the physics.
- We prepare its ground state by **VQE** and read the topological signature from a circuit-measurable observable.

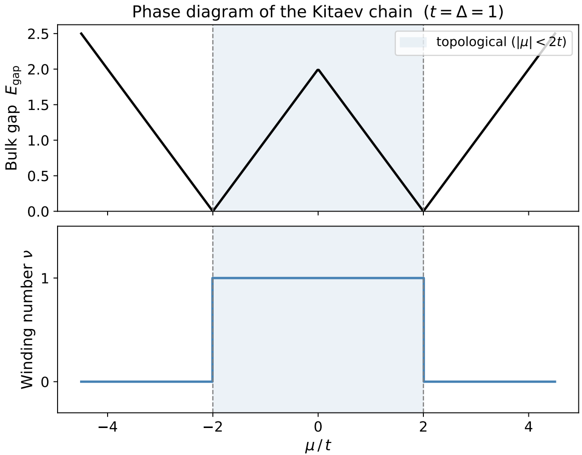
The question

Can a *topological signature* be prepared and still *survive* on a noisy NISQ device — and how far in system size L can it be pushed?

Four Blocks:

1. the physics bridge,
2. finite-size physics & qubit encoding,
3. measuring topology with circuits,
4. the NISQ reality check (results).

Block 1 — The Physics Bridge

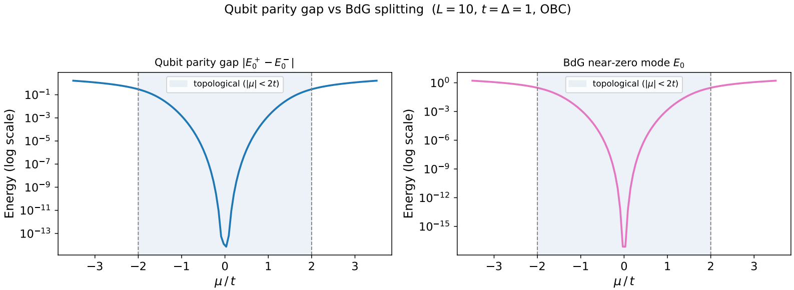


The Kitaev chain: p-wave pairing on a 1D lattice,

$$E_k = \sqrt{(\mu + 2t\cos k)^2 + (2\Delta \sin k)^2}.$$

- **Bulk gap** closes only at $|\mu| = 2t$ — the phase boundary.
- **Topological phase** for $|\mu| < 2t$: a $\mathbb{Z}_2$ **winding number** $\nu = 1$; trivial $(\nu = 0)$ outside.
- Bulk–boundary correspondence $\Rightarrow$ **Majorana zero modes** pinned to the ends in the topological phase.

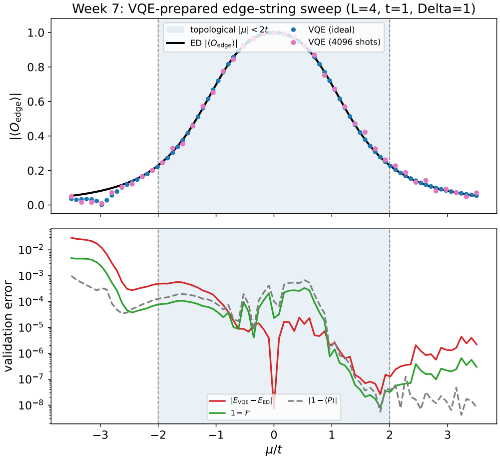
Block 2 — Finite-Size Physics & Qubit Encoding



On a finite chain the two end Majoranas overlap and split, $\Delta_P(L) \sim e^{-L/\xi}$.

- **Exponential protection**: the parity gap shrinks as the chain grows — longer is better.
- **Jordan–Wigner** gives the *exact* qubit Hamiltonian $H = -\frac{\mu}{2}\sum_j Z_j + \frac{t-\Delta}{2}\sum_j X_j X_{j+1} + \frac{t+\Delta}{2}\sum_j Y_j Y_{j+1}$.
- **Parity** $\hat{P} = \prod_j Z_j$ is a $\mathbb{Z}_2$ symmetry: even/odd sectors label the two near-degenerate ground states.

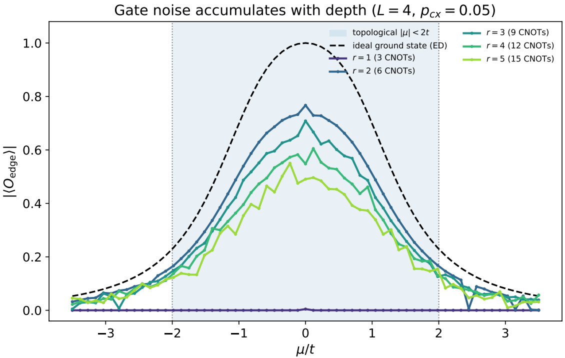
Block 3 — Measuring Topology with Quantum Circuits (VQE)

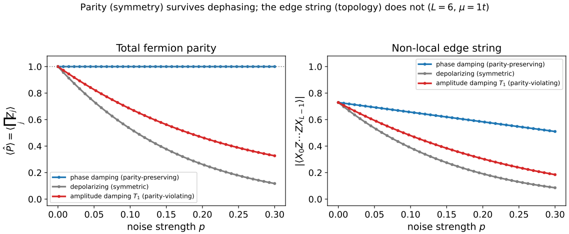


Prepare the ground state *variationally*, then detect the phase from a *circuit-measurable* observable.

- **Parity-penalized VQE** (`EfficientSU2`) prepares the even-sector ground state — no exact eigenvector needed.
- Topology lives in a **non-local edge string** $O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$.
- Sweeping μ: the string **turns on** in the topological phase, off in the trivial — it tracks the transition.

Block 4 — NISQ Reality Check I: Noise Breaks the Diagnostic

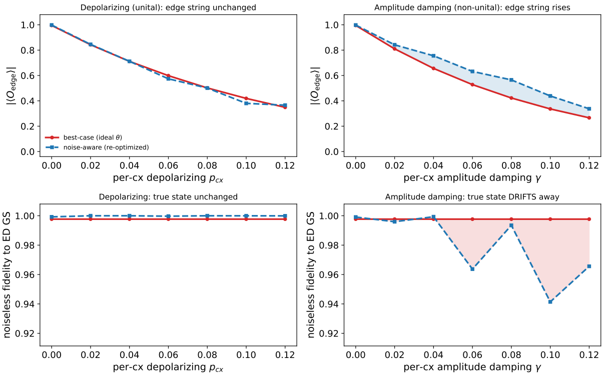




Circuit-level gate noise enters *through the gates*: each cx adds an error channel, so the signal decays as $(1 - p_{\mathrm{cx}})^{r(L-1)}$ with depth. **Left:** an under-expressive ansatz is dead everywhere; once expressive, deeper circuits are pushed down by accumulated CNOT noise. **Right:** parity is a symmetry (protected under dephasing), but the edge string (topology) is not noise-immune — it decays under every channel.

Block 4 — NISQ Reality Check II: Optimizing $\neq$ Preparing

Optimizing the noisy cost is not preparing the state: non-unital noise inflates the edge string while the true-state fidelity falls ($L = 4$, $r = 3$, $\mu = 0$, $\lambda = 0.1$)



With a NoiseModel-backed estimator, *every* VQE cost evaluation is corrupted.

- **The two rows:** *top* = measured edge string; *bottom* = **true-state fidelity** to the exact ground state.
- Non-unital noise (T_1, *right*) **inflates the edge string** while **fidelity collapses** — the observable lies.
- Optimizing the noisy cost $\neq$ preparing the state.

Block 4 — NISQ Reality Check III: Scaling to Large L (Methods)

Past the dense $L \approx 15$ wall: remove every exponential object but the mild 2^L statevector.

Removing the two exponentials

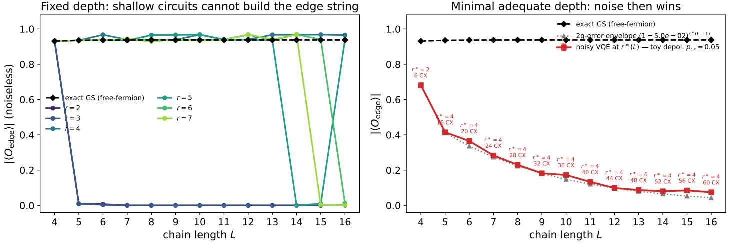
- **ED $\rightarrow$ free fermions**: the model is quadratic — exact GS & edge string from a $2L \times 2L$ **Bogoliubov–de Gennes** matrix, $O(L^3)$.
- 4^L **density matrix $\rightarrow$ MPS trajectories**: each noisy shot is one **matrix-product state (MPS)** trajectory — a chain of per-site $\chi \times \chi$ tensors with bond dimension $\chi \leq 2^r$ ($= 16$ at $r^* = 4$), so memory is $O(L\chi^2)$: **linear in** L, not 4^L.

The mild object + HPC

- **Ideal VQE**: a 2^L statevector with *sparse* Pauli operators — never the 4^L dense H ($\sim 500\times$ faster at $L = 14$).
- $\mu = 0.5t$ (gapped), L-BFGS-B, ED-free even-sector selection.
- (L, reps) grid parallelized on the **Leipzig SC cluster**.

Block 4 — NISQ Reality Check IV: The Scaling Result

Two ways a fixed-depth preparation fails as L grows ($\mu = 0.5t$, noise: toy depol. $p_{cx} = 0.05$)

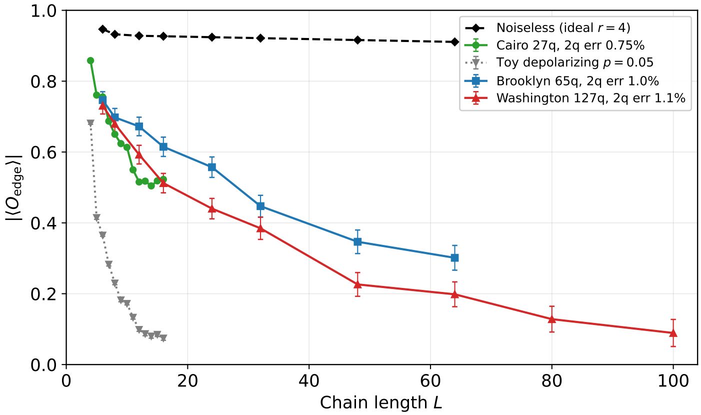


The two failure modes decouple.

- **Left:** expressibility is *cheap and L-independent* — a **constant** depth $r^* = 4$ reproduces the edge string to $L = 16$ (shallow $r = 2, 3$ fail).
- **Right:** at fixed depth $n_{\mathrm{cnot}} = 4(L-1)$ grows *linearly*, so **noise is the sole large-L wall** — the edge string decays exponentially.

Toy ($p = 0.05$): dead by $L \approx 14$. Real Cairo ($2q$ err ≈ 0.0075, $\sim 6\times$ gentler) keeps $\langle O_{\mathrm{edge}} \rangle$ *plateauing* ~ 0.5 to $L = 16$ — it **survives far further** (next slide).

Block 4 — NISQ Reality Check V: Real Hardware to $L = 100$



Every run at $r^* = 4$, one axis.

- **Noiseless** (black): flat $\sim$0.9 — topological at *every* L.
- **Toy** p=0.05: $\sim$6$\times$ too harsh, dead by $L = 16$.
- **Real IBM** (Cairo / Brooklyn / Washington): survive far further; **agree on the overlap**, ordered by 2q error.
- Gone ($\sim$0.09) only by $L = 100$ — decoherence over a *linear* CNOT count.

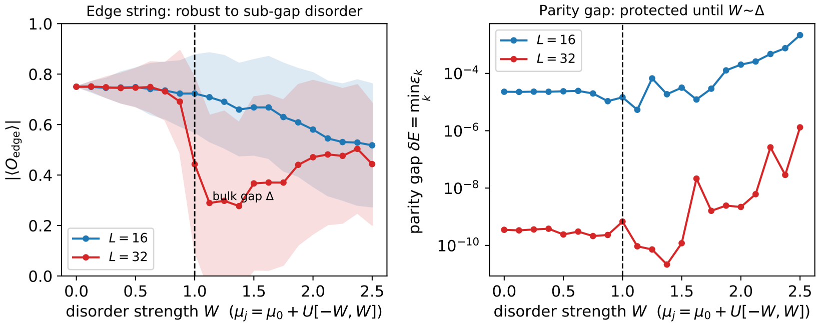
Block 4 — Which Physical Errors Kill the Topology?

A perturbation is harmless only if it is **(i) local**, **(ii) fermion-parity preserving**, and **(iii) weaker than the bulk gap** Δ. Break any one and the phase dies.

Physical error (Kitaev H)	**Jordan–Wigner $\rightarrow$ qubit**	**Verdict**
Charge noise $\delta\mu_i\, c_i^\dagger c_i$	$\sum_i \delta h_i\, Z_i$ (parity-even)	**robust** (up to Δ)
Quasiparticle poisoning $c_i^\dagger$	$\left(\prod_{k<i} Z_k\right)(X_i \mp iY_i)$ (parity-*odd*)	**fatal**: flips the bit
Gap closure $\mu \to 2t$	$\sum_i Z_i$ dominates $\sum_i X_i X_{i+1}$	**fatal**: phase transition

What to measure: the string order O_{edge}, the fermion parity $P = \prod_j Z_j$, and the splitting $\delta E = |E_{\mathrm{even}} - E_{\mathrm{odd}}| \sim e^{-L/\xi}$.

Block 4 — Robust Error: Local Disorder (Exact, to $L = 32$)

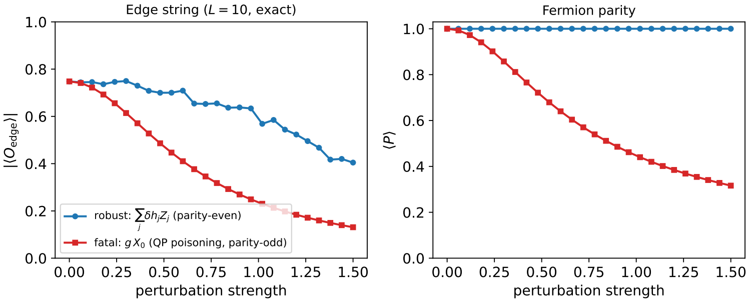


Charge noise $=$ a random field $\sum_j \delta h_j Z_j$ (parity-even): per-site $\mu_j = \mu_0 + \mathrm{Uniform}(-W, W)$, where W is the disorder half-width (the x-axis). Free-fermion BdG, exact.

- Below the bulk gap Δ the edge string **holds** and the splitting stays *exponentially* small.
- Past $W \sim \Delta$ the gap closes and both die.

Local, parity-even, sub-gap $\Rightarrow$ topologically harmless.

Block 4 — Fatal Error: Quasiparticle Poisoning

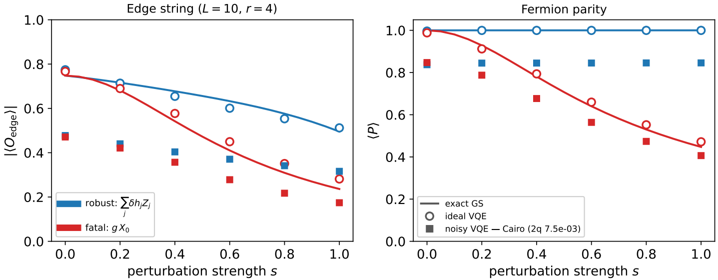


A stray electron $c_0 + c_0^\dagger = \gamma_0 = X_0$ is **parity-odd**. Exact diagonalisation, $L = 10$.

- Parity-even Z-disorder: $\langle P \rangle = +1$, edge string survives.
- Parity-odd $g\,X_0$: $\langle P \rangle$ and O_{edge} **collapse**.

QP poisoning is the Achilles heel: the poisoning time must beat the gate time.

Block 4 — The Physical Error on the Real Circuit (VQE)



The taxonomy on the *actual* VQE circuit (L=10, r=4): ideal statevector (∘) and noisy Cairo density matrix (■) over the exact curves.

- Ideal VQE **tracks the exact ground truth** for both channels — the qubit ansatz reproduces the physics.
- Robust Z-disorder: $\langle P\rangle$ stays $+1$; fatal $g\,X_0$ **collapses** $\langle P\rangle$ — the parity breakdown *on the qubits*.
- Cairo noise stacks *below* the ideal markers: physical error × gate error.

Block 4 — Device Noise *is* Physical Error (Jordan–Wigner)

Two axes — **physical** errors perturb the *simulated wire*; **NISQ** errors perturb the *circuit* that simulates it — but Jordan–Wigner sends every single-qubit channel to a Pauli, so the *same* parity rule classifies both.

NISQ error	**Physical-error twin**	**Verdict**
T_2 dephasing ($\sim Z_i$)	charge noise $\delta\mu\, n_i$	parity-even: robust
coherent Z miscalibration	coherent $\delta\mu$ shift	robust
T_1 damping ($\sigma^- = \frac{1}{2}(X + iY)$)	**quasiparticle poisoning**	parity-odd: fatal
depolarising ($X + Y + Z$)	$\frac{2}{3}$ poisoning, $\frac{1}{3}$ charge noise	fatal component

So T_1 relaxation *is* poisoning and T_2 dephasing *is* charge noise — not analogies, the same operators. **But** the gap protects against physical errors only when the *hardware is the wire*; our transmons are *not* the wire, so a mid-circuit X is unprotected: **the taxonomy transfers, the protection does not**.

Takeaways

- The Kitaev chain's Majorana edge signature can be **prepared and measured on qubits** via a non-local string order $O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$.
- **Parity (symmetry) is protected** under noise; the **edge string (topology) is not** noise-immune at fixed L.
- Both exponentials are classically removable: ED $\rightarrow$ **free fermions** (BdG), the 4^L density matrix $\rightarrow$ **MPS trajectories** — only a mild 2^L statevector remains.
- **Expressibility is L-independent** (constant depth $r^* = 4$): the large-L wall is *decoherence* over a linearly growing gate count.
- On **real IBM noise** the signature survives far beyond the toy model ($p = 0.05$), staying measurable to $L \sim 48$ across devices.
- **Which errors kill topology?** Local parity-even sub-gap noise is harmless; **quasiparticle poisoning** (parity-odd) and gap closure are fatal — and via JW the device's own T_1 *is* poisoning, T_2 *is* charge noise.

Notes: `scaling_to_large_L.tex`, `kitaev_error_taxonomy.tex`, `parity_topology_protection.tex`.